Department of Statistics and Data Science
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Lecture 4: From Counting Theory to Discrete Random Variables

Ordering, Arrangement, Combination, Concrete PMFs, Uniform, Bernoulli, and Binomial Models

Nurturing Future Risk Managers and Actuaries:
An AI-powered Journey for Gifted Secondary Students

Purpose of this set of notes. These notes are designed for self-learning. They merge the counting structures needed to describe sample spaces with the discrete probability models used to place weights on those sample spaces. The goal is not merely to compute answers, but to make the logical architecture of risk models visible. Each section follows a gifted-learner structure: a cognitive trigger, a structural core, proof decision points, a deepening layer, and an extension layer.

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How to Use This Document

Structural Core

Core Read every section in the order given. The chapter is built so that counting provides the structural skeleton for probability.

Insight Do not rush to formulas. Always decide first whether you are counting *outcomes*, describing a *random variable*, or attaching *probability weights*.

Extension Whenever you see a Gemini prompt, use it as a launch-pad for self-directed inquiry rather than as an answer machine.

Pause-and-Think

For each problem you attempt, ask yourself three questions before calculating:

1. What exactly is being counted or measured?
2. Does order matter?
3. Is the problem purely combinatorial, or does it already involve probability?

Module I

Counting as the architecture of discrete risk

1 Why Counting Comes Before Probability

Cognitive Trigger

A transport insurer sells a policy that pays if a flight is delayed, but the operational risk team is worried about something deeper than a single delay. They want to understand the *space of scenarios*: which passengers may claim, in which order claims arrive, and how many ways a system can be stressed. Why is the number of possible scenarios already a form of risk, even before any probabilities are assigned?

Structural Core

Big structural idea. In a discrete setting, probability is always attached to a *sample space*. Counting tells us how large the space is; probability describes how weight is distributed across it. If the space is misunderstood, every later probability calculation will rest on the wrong foundation.

Proof Decision Point

When a problem feels complicated, your first proof-decision is not "Which formula do I use?" but rather "What is the object I am distinguishing from everything else?"

- A **path**? Then stages matter.
- A **selection**? Then membership matters, not order.
- A **coded output**? Then repetition rules matter.
- A **later random variable**? Then counting will often control the coefficient of a probability expression.

Deepening Layer

Deepening question. Suppose two systems both have 120 possible end states. In System A, each end state is built through only two stages. In System B, each end state is built through six stages. Which system is more operationally fragile, even if the total count is the same? Explain why counting alone may be insufficient unless we also understand structure.

1.1 The Fundamental Counting Principle (FCP)

Structural Core

If a task is completed in several successive stages, and the number of available choices at each stage is independent of the others, then

$$N = n_1 \times n_2 \times \cdots \times n_k.$$

This is the **product principle**. If instead a task can be done by one of several *non-overlapping alternatives*, then the total number of possibilities is the sum of the counts of those alternatives. This is the **sum principle**.

Worked / Contrast Example

Paired example: what works and what fails.

- **Working case.** A loan approval pipeline has 5 automated scores, 3 manual review ratings, and 2 rider options. The complete approval path count is $5 \times 3 \times 2 = 30$.
- **Failing case.** A student says that choosing one hedge from 8 futures or 5 options should also be 8×5. This fails because the decision is *either/or*, not *one stage followed by another*.

Critical distinction: multiplication builds a sequence of choices; addition merges exclusive branches.

Critical Condition / Common Trap

A common error is to multiply whenever several numbers appear in the problem statement. The correct question is not "How many numbers are given?" but "Do these choices happen together as stages, or separately as alternatives?"

Proof Decision Point

Proof decision point for the product principle. The key logical move is to think of each final outcome as a unique *ordered tuple* $(a_1, a_2, \ldots, a_k)$. Once you represent each complete outcome by a tuple of stage-choices, multiplication becomes unavoidable. If you cannot represent the outcomes as tuples, you should be suspicious that multiplication may not be the right model.

Deepening Layer

Why the rule works. The product principle is really a statement about constructing a bijection between complete outcomes and tuples of choices. This is one of the most important hidden ideas in combinatorics: we count hard objects by matching them to simpler objects that are easier to count.

For a deeper dive, try this with Google Gemini

Use this exact prompt:

> "Explain the Fundamental Counting Principle using three different metaphors: one from music composition, one from computer passwords, and one from logistics planning. For each metaphor, identify when multiplication is valid and when addition is valid. Include one counterexample where students are tempted to multiply but should not."

1.2 Factorials as the Logic of Exhaustion

Cognitive Trigger

Seven companies in a stressed bond portfolio eventually default. The set of defaulting companies is known, but the *order of default* is not. Why does this timeline question explode far faster than intuition expects?

Structural Core

For a positive integer n,

$$n! = n(n-1)(n-2)\cdots 2 \cdot 1,$$

with the convention $0! = 1$. Factorials arise when every stage uses up one previously available option.

Proof Decision Point

Proof decision point for factorial structure. The essential choice is to focus on the *first position*. There are n possibilities for the first slot, then $n-1$ for the second, then $n-2$, and so on. This recursive peeling-off strategy is a signature move in combinatorics: reduce the full problem by fixing an early choice and counting what remains.

Worked / Contrast Example

Paired example.

- **Working case.** Arranging all 7 distinct bonds by order of default gives $7! = 5040$ possible sequences.
- **Failing case.** A student claims that if two bonds are operationally indistinguishable, the answer is still $7!$. This fails because swapping indistinguishable objects does not create a genuinely new arrangement.

Deepening Layer

Why is $0! = 1$ reasonable? The convention is not arbitrary bookkeeping. It preserves structural consistency. If $n!$ counts all arrangements of n objects, then the number of ways to arrange *nothing* is the number of ways to leave the arrangement unchanged: exactly one. More deeply, formulas like $\binom{n}{0} = 1$ collapse without this convention.

Extension Layer

Generalisation challenge. Consider an algorithm that explores all possible sequences of n actions. Why is saying "the runtime is factorial" substantially more alarming than saying "the runtime is exponential"? Research how quickly $n!$ overtakes 2^n, and try to explain the reason without merely comparing tables of values.

For a deeper dive, try this with Google Gemini

Use this exact prompt:

> "Act as a mathematician coaching a gifted secondary student. Do not prove factorial identities fully. Instead, list the key decision points needed to justify why $n!$ counts arrangements of n distinct objects and why $0! = 1$. Then propose two unusual applications outside pure mathematics."

1.3 Combinations: membership matters

Cognitive Trigger

An audit team selects 3 bonds from 10 for inspection. Why should the answer be much smaller than the number of ways to inspect the same three in a first-second-third sequence, and what exactly is being "divided away"?

Structural Core

The number of unordered selections of r objects from n distinct objects is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

The denominator records two kinds of overcounting:

- $r!$: different orderings of the chosen group,
- $(n-r)!$: different orderings of the unchosen group when starting from $n!$.

Proof Decision Point

Proof decision point. The key move is to notice that every selected group of size r appears exactly $r!$ times among the ordered selections of those same r objects. Once you recognise this multiplicity, division is no longer a trick; it becomes a necessity.

Worked / Contrast Example

Paired example.

- **Working case.** Selecting 3 bonds from 10 gives $\binom{10}{3} = 120$.

Deepening Layer

Symmetry as structure. Why is $\binom{n}{r} = \binom{n}{n-r}$? Instead of proving it algebraically first, explain it in words: selecting the r objects to keep is equivalent to selecting the $n-r$ objects to leave out. This symmetry is a powerful diagnostic tool; it often helps you detect impossible answers quickly.

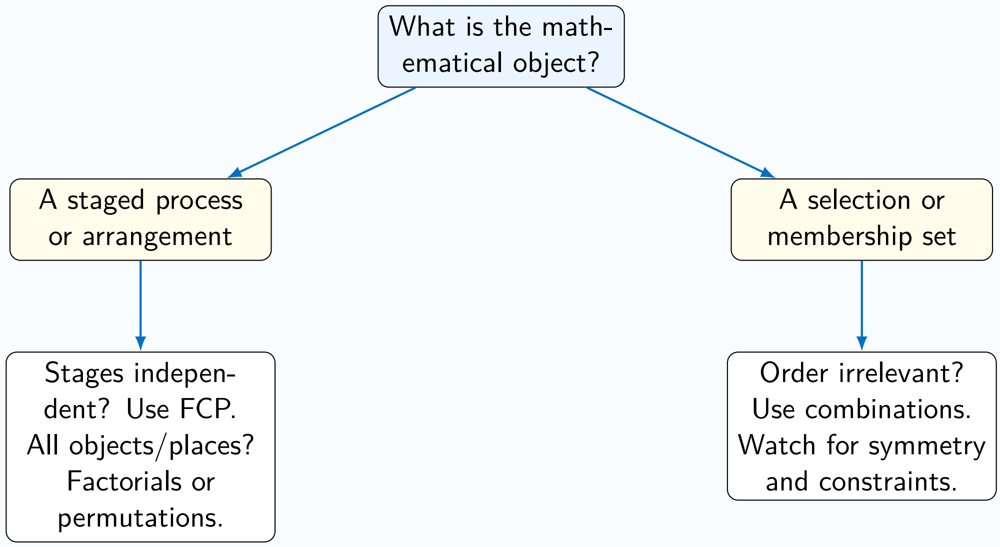
For a deeper dive, try this with Google Gemini

Use this exact prompt:

> "Explain the formula for $\binom{n}{r}$ without giving a full proof. Focus on the decision points: what is overcounted, why overcounting occurs, and why the division by $r!$ is structurally necessary. Then design three counterexamples that help students decide whether a problem is a permutation or a combination."

2 Counting Summary Map

Synthesis



Pause-and-Think

Self-check before moving on.

1. Why is counting not yet probability?
2. How do factorials embody "decreasing availability"?
3. What is the exact overcount removed when moving from permutations to combinations?
4. Why is fixing one object legitimate in circular arrangements but not in linear ones?

Module II

From counted outcomes to discrete random variables

3 Random Variables: turning outcomes into numbers

Cognitive Trigger

A prize wheel has sectors labelled \$0, \$10, and \$50, but the wheel itself does not care about money. The money values are assigned by the organiser. Where, exactly, does the mathematics of a random variable begin: in the wheel, in the outcomes, or in the rule that translates outcomes into numbers?

Structural Core

A **random variable** is a function from outcomes in a sample space to numerical values. In a **discrete** model, the set of possible values is countable. The probability mass function (PMF) is the rule

$$\mathrm{P}(X = x)$$

that assigns a probability to each admissible value x, subject to two conditions:

1. $0 \le \mathrm{P}(X = x) \le 1$ for every possible value x,
2. $\sum_x \mathrm{P}(X = x) = 1$.

Proof Decision Point

Proof decision point. The fundamental move is to separate three levels:

1. the underlying outcomes,
2. the values assigned by the random variable,
3. the probabilities bundled together when different outcomes map to the same value.

Many conceptual errors happen because students jump directly from a story to PMF values without asking what distinct outcomes collapse to the same numerical value.

Worked / Contrast Example

Working case. Let X be the payout of a promotional wheel with $\mathrm{P}(X = 50) = 0.05$, $\mathrm{P}(X = 10) = 0.20$, and therefore $\mathrm{P}(X = 0) = 0.75$.

- The values of X are $\{0, 10, 50\}$.
- The PMF is valid because probabilities are non-negative and sum to 1.

Failing case. If someone writes $\mathrm{P}(X = 0) = 0.80$ without changing the other two probabilities, the total becomes 1.05, so the PMF fails the summation condition.

Deepening Layer

Why the PMF condition matters. The equation $\sum_x \mathrm{P}(X = x) = 1$ is not just a checklist. It expresses completeness: every possible value must be accounted for, and no probability mass can go missing or appear twice. When students later study continuous models, this same idea reappears as "total area under a density equals 1."

For a deeper dive, try this with Google Gemini

Use this exact prompt:

> "Explain the difference between an outcome, an event, and a random variable using one casino example and one insurance example. Then give a short challenge where two different outcomes produce the same random-variable value, and ask the student to compute the PMF."

3.1 Validity, support, and cumulative probability

Structural Core

Support means the set of all values that a random variable can actually take.
Cumulative probability means combining PMF values over a range, for example

$$\mathrm{P}(X \leq 2) = \mathrm{P}(X = 0) + \mathrm{P}(X = 1) + \mathrm{P}(X = 2).$$

Discrete probability often looks simple because the support can be written down explicitly, but the thinking is subtle: every range probability is a controlled sum over exactly the values included.

Critical Condition / Common Trap

Students often compute "at most" or "at least" using the wrong endpoint. "At most 2" means $0, 1, 2$ if those are the relevant values, whereas "less than 2" means only values strictly below 2. Boundary language matters.

4 Discrete Random Variables Defined by Concrete Probabilities

Cognitive Trigger

Two insurers describe risk differently. Company A says, "There are 120 possible claim scenarios." Company B says, "There are 120 scenarios, but most probability mass is concentrated in just 10 of them." Which company actually understands risk better, and why is counting alone insufficient here?

Structural Core

A discrete random variable is not fully defined by its possible values alone. It is defined by:

- its **support**: the set of possible values,
- its **probability mass function**: the specific probability attached to each value.

A distribution is **concrete** when $\mathrm{P}(X = x)$ is explicitly given for each admissible value x.

Proof Decision Point

Critical modelling decision. When given a distribution, ask:

- Are the probabilities derived from **symmetry**?
- Are they derived from **data**?
- Are they produced by an **algebraic rule**?

The same support can support radically different models depending on how the mass is assigned.

Worked / Contrast Example

Paired example.

x	0	1	2
$\mathrm{P}(X = x)$	0.5	0.3	0.2

This is a valid PMF because all probabilities are non-negative and sum to 1.
Failing partner example.

x	0	1	2
$\mathrm{P}(X = x)$	0.5	0.3	0.4

This fails because the total is 1.2, so the model is not a probability distribution.

Critical Condition / Common Trap

A frequent misconception is: "If a variable has three values, each must have probability $\frac{1}{3}$." That is true only in the uniform case. In most real contexts, unequal weighting is exactly what makes the model informative.

Deepening Layer

Deep structure question. If two random variables have the same support but different PMFs, what changes and what stays the same? Try to separate:

- what the variable *can* do,
- what the variable *usually* does,
- and what the variable *rarely but importantly* does.

This is the beginning of thinking like a risk modeller.

For a deeper dive, try this with Google Gemini

Use this exact prompt:

> "Explain the difference between a sample space and a probability distribution using one example where all outcomes are equally likely and one where probabilities are highly unequal. Then create a challenge where two different PMFs share the same support but lead to very different expectations."

5 General Probability Mass Functions Defined by Formula

Cognitive Trigger

Suppose a model is specified not by a table but by the rule

$$\mathrm{P}(X = x) = k(x^2 + 1), \qquad x \in \{0, 1, 2, 3\}.$$

Where does the actual distribution come from, and why is the constant k not a cosmetic extra but a structural necessity?

Structural Core

A **functional PMF** defines probabilities through an algebraic rule. To turn the rule into an actual probability distribution, we impose the normalisation condition

$$\sum_x \mathrm{P}(X = x) = 1.$$

This determines any unknown constants.

Proof Decision Point

Proof decision point. The important logical sequence is:

1. list every allowed value of x,
2. substitute into the formula,
3. add all resulting expressions,
4. set the total equal to 1.

In other words, algebra does not replace probability logic; it must submit to it.

Worked / Contrast Example

Working case.

$$\mathrm{P}(X = x) = k(x^2 + 1), \qquad x = 0, 1, 2, 3.$$

Then

$$\mathrm{P}(0) = k, \quad \mathrm{P}(1) = 2k, \quad \mathrm{P}(2) = 5k, \quad \mathrm{P}(3) = 10k.$$

So

$$k + 2k + 5k + 10k = 18k = 1, \qquad k = \frac{1}{18}.$$

Failing partner example. A student sets $k = 1$ simply because the formula looks complete. This ignores total probability and produces an impossible distribution.

Deepening Layer

Deeper layer. A functional PMF separates **shape** from **scale**. The expression $(x^2 + 1)$ suggests how the mass grows with x, while the constant k rescales the whole structure so that the total mass is exactly 1. This idea returns later in continuous probability through density normalisation.

Extension Layer

Try inventing three different formula-based PMFs on the same support $\{0, 1, 2, 3\}$: one increasing, one decreasing, and one symmetric. How much freedom do you really have once non-negativity and total probability are enforced?

For a deeper dive, try this with Google Gemini

Use this exact prompt:

> "Give three examples of probability mass functions defined using algebraic formulas. For each, explain how to determine the normalising constant and describe the qualitative shape of the distribution (increasing, decreasing, symmetric)."

6 Discrete Uniform Distribution: when counting becomes probability

Cognitive Trigger

A student rolls a fair die and a biased die. Both have six possible outcomes. Why does counting give the same number of possible outcomes in both cases, but probability behaves completely differently?

Structural Core

A discrete random variable X is **uniformly distributed** over n admissible values if

$$\mathrm{P}(X = x) = \frac{1}{n} \quad \text{for every admissible } x.$$

This is the unique setting in which counting directly yields probability:

$$\mathrm{P}(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{number of equally likely outcomes}}.$$

Proof Decision Point

Modelling decision. Before using a uniform model, ask two separate questions:

- Is the situation genuinely symmetric?
- Or am I using uniformity merely because I lack better information?

These are not the same. One is a model justified by structure; the other is a provisional baseline.

Worked / Contrast Example

Paired example. For a fair die,

$$P(X = x) = \frac{1}{6}, \qquad x \in \{1, 2, 3, 4, 5, 6\}.$$

So the probability of an even number is

$$\frac{3}{6} = \frac{1}{2}.$$

Failing partner example. For a biased die, there are still three even faces, but the probability is no longer $\frac{3}{6}$ unless those faces retain equal total mass. Counting by itself is no longer enough.

Critical Condition / Common Trap

Students often overgeneralise the rule

$$\frac{\text{favourable}}{\text{total}}.$$

This is not the definition of probability in general. It is a special shortcut valid only under uniformity.

6.1 Expectation via symmetry

Structural Core

If $X \sim \text{Uniform}(\{1, 2, \ldots, n\})$, then

$$E[X] = \frac{n+1}{2}.$$

This formula is best understood as a symmetry statement, not a memory item.

Proof Decision Point

Proof decision point. Pair the outer values:

$$1 + n, \qquad 2 + (n - 1), \qquad 3 + (n - 2), \ldots$$

Every pair has the same sum, so the centre of balance must be $\frac{n+1}{2}$. The expectation emerges from structural symmetry before any heavy algebra is needed.

Worked / Contrast Example

For a fair 12-sided die,

$$P(X = x) = \frac{1}{12}, \qquad x = 1, 2, \ldots, 12.$$

The probability of a multiple of 3 is

$$\frac{4}{12} = \frac{1}{3}$$

because the favourable values are $3, 6, 9, 12$. The expected value is

$$E[X] = \frac{1 + 12}{2} = 6.5.$$

Structural insight. The event probability comes from counting because the model is uniform; the expectation comes from symmetry because the support is evenly spaced.

6.2 Variance and spread in the uniform model

Structural Core

If $X \sim \text{Uniform}(\{1, 2, \ldots, n\})$, then

$$\text{Var}(X) = \frac{n^2 - 1}{12}.$$

This measures how widely the equally likely values are spread around the centre.

Deepening Layer

Deepening question. Compare a fair coin coded as $\{0, 1\}$, a fair die coded as $\{1, 2, 3, 4, 5, 6\}$, and a fair 20-sided die coded as $\{1, 2, \ldots, 20\}$. All are uniform, but their variances are very different. Which feature of the support, not the equal weighting itself, controls this increasing spread?

6.3 Uniform distribution as a baseline model

Deepening Layer

Uniform distributions are often used as baseline models in risk work:

- as a neutral stress test,
- as a starting assumption before richer data are available,
- as a deliberately maximally non-committal model over a fixed finite set.

But a baseline model is not automatically a good predictive model. That distinction matters in actuarial judgement.

Extension Layer

Compare the uniform and binomial worlds conceptually:

- In the uniform model, every admissible outcome is weighted equally.
- In the binomial model, outcomes with the same count of successes are grouped together, and their probabilities are shaped by both counting and the success probability p.

In other words, the move from uniform to binomial is a move from symmetry-based probability to structure-based probability.

For a deeper dive, try this with Google Gemini

Use this exact prompt:

> "Compare discrete uniform distribution and binomial distribution in terms of how probabilities are assigned. Include three real-world examples where uniform is appropriate and three where it leads to misleading results. Do not compute anything; focus only on modelling assumptions."

7 Bernoulli Distribution: the atomic risk model

Cognitive Trigger

A borrower either defaults or does not default. A patient either survives the year or does not. Why does such a tiny two-outcome model deserve to be called the atom of risk modelling?

Structural Core

A random variable X follows a Bernoulli distribution with parameter p, written $X \sim \mathrm{Bern}(p)$, if

$$X \in \{0, 1\}, \qquad \mathrm{P}(X = x) = p^x (1-p)^{1-x}.$$

Here $X = 1$ represents the event of interest (often called "success"), and $X = 0$ represents its complement.

Proof Decision Point

Proof decision point. Why does the compact formula work? Because the exponents act as switches:

- when $x = 1$, the formula becomes p,
- when $x = 0$, the formula becomes $1 - p$.

The insight is not the formula itself, but the idea that algebra can encode case-splitting elegantly.

Worked / Contrast Example

Insurance mortality example. If the one-year mortality probability is $p = 0.0015$, then for $X = 1$ meaning "death within one year,"

$$\mathrm{P}(X = 1) = 0.0015, \qquad \mathrm{P}(X = 0) = 0.9985.$$

If the benefit is \$1,000,000 paid on death, the expected payout is

$$1{,}000{,}000 \times 0.0015 = 1500.$$

Interpretation: even a very unlikely event can generate a substantial expected cost when the severity is large.

Deepening Layer

Deep structure. Bernoulli models are simple not because they are trivial, but because they isolate the basic idea of uncertainty in one trial. Every later frequency model in these notes essentially asks: what happens when we combine many Bernoulli trials?

Extension Layer

Suppose the "success probability" is not fixed, but changes over time because of climate, regulation, or contagion. Which part of the Bernoulli model breaks first: the support $\{0, 1\}$, the PMF formula, or the interpretation of p? Explain your answer.

For a deeper dive, try this with Google Gemini

Use this exact prompt:

> "Pretend you are an actuary explaining Bernoulli trials to a gifted secondary student. Give two analogies: one from epidemiology and one from cybersecurity. Then challenge the student with examples where the outcome is binary but the success probability is not constant over time. Do not solve the challenge."

7.1 Expectation and variance of a Bernoulli variable

Structural Core

For $X \sim \mathrm{Bern}(p)$,

$$\mathrm{E}[X] = p, \qquad \mathrm{Var}(X) = p(1 - p).$$

The mean equals the probability of the event; the variance depends on both the event and its complement.

Proof Decision Point

Why the variance has the factor $1 - p$. Uncertainty is smallest when one outcome is nearly certain. The factor $p(1-p)$ reflects this symmetry: uncertainty disappears both when $p \to 0$ and when $p \to 1$, and peaks in the middle.

8 Binomial Distribution: aggregating Bernoulli risk

Cognitive Trigger

A travel insurer covers 2,000 passengers, each with a 2.5% chance of a qualifying delay. Even if a single delay is a Bernoulli event, why is the total number of delayed passengers not Bernoulli? What extra structure appears the moment we aggregate?

Structural Core

A random variable X follows a binomial distribution with parameters n and p, written $X \sim \mathrm{Bin}(n, p)$, if it counts the number of successes in n independent Bernoulli trials with common success probability p. Its PMF is

$$\mathrm{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, 2, \ldots, n.$$

A useful checklist is **BINS**:

- **B**: Binary outcomes,
- **I**: Independent trials,
- **N**: Fixed number of trials,
- **S**: Same success probability.

Proof Decision Point

Proof decision point. The binomial formula has two components with different meanings:

- $p^x(1-p)^{n-x}$ gives the probability of one *specific* sequence with exactly x successes,
- $\binom{n}{x}$ counts how many such sequences exist.

The formula is therefore a perfect fusion of the two modules of this chapter: counting *times* probability.

Worked / Contrast Example

Home-owner fire claims. If 15 policies are independent and each has fire-claim probability 0.02, then for exactly 2 claims,

$$\mathrm{P}(X = 2) = \binom{15}{2} (0.02)^2 (0.98)^{13}.$$

The combination term counts *which* 2 policies claim. The powers of 0.02 and 0.98 encode the probabilistic weight of any one such pattern.

Critical Condition / Common Trap

A common mistake is to think that $\binom{n}{x}$ is a probability. It is not. It is a count. The PMF becomes a probability only after the count is multiplied by the probability of a single success/failure pattern.

Deepening Layer

Why independence matters. In the 2008 financial crisis, many defaults were not independent. Once trials become linked, the elegant binomial structure begins to distort reality. In other words, the formula may still produce a number, but the number may describe the wrong world.

Extension Layer

Deep question. Suppose students know only the formula $\binom{n}{x}p^x(1-p)^{n-x}$, but have forgotten the story behind it. Can they reconstruct the meaning of the model from the formula alone? Challenge yourself to infer what each factor must represent before reading any explanation.

For a deeper dive, try this with Google Gemini

Use this exact prompt:

> "Teach the binomial distribution to a gifted student by revealing the formula piece by piece. First explain what one fixed success/failure sequence contributes. Then explain why a counting coefficient is needed. Finally, create one example where the BINS conditions hold and one example where independence fails badly. Do not provide full solutions."

8.1 Useful binomial probabilities

Structural Core

For binomial models, many practical questions involve cumulative probabilities:

- $\mathrm{P}(X \le k)$: "at most k successes",
- $\mathrm{P}(X \ge k)$: "at least k successes",
- complements such as $\mathrm{P}(X \ge 1) = 1 - \mathrm{P}(X = 0)$.

In practice, complement methods often simplify large calculations.

Worked / Contrast Example

For the same 15-policy fire model,

$$\mathrm{P}(X \ge 1) = 1 - \mathrm{P}(X = 0) = 1 - (0.98)^{15}.$$

This is easier than adding $\mathrm{P}(X = 1) + \mathrm{P}(X = 2) + \cdots + \mathrm{P}(X = 15)$. The strategic skill is not just to compute, but to select the direction of attack that collapses the work.

8.2 Expectation and variance of the binomial distribution

Structural Core

If $X \sim \text{Bin}(n, p)$, then

$$\text{E}[X] = np, \qquad \text{Var}(X) = np(1-p), \qquad \sigma = \sqrt{np(1-p)}.$$

Interpretation in risk work:

- np is the average number of claims, defaults, or failures,
- $np(1-p)$ measures spread and therefore impacts reserve planning and capital buffers.

Proof Decision Point

Why does the variance peak near $p = 0.5$? Because uncertainty is largest when success and failure are most balanced. When one side dominates, the count becomes more predictable. This is a conceptual explanation worth understanding before any calculus is introduced.

Worked / Contrast Example

If a cyber insurer covers 1,000 firms with breach probability 0.04, then

$$\text{E}[X] = 1000(0.04) = 40, \qquad \sigma = \sqrt{1000(0.04)(0.96)} \approx 6.20.$$

The mean tells the insurer what is typical; the standard deviation tells the insurer how far a real year may drift from the mean.

9 Synthesis: four ways discrete distributions arise

Synthesis

The discrete distributions in this study set arise in four structurally different ways:

Type	Origin	Example
Uniform	symmetry or neutral baseline	fair die
Concrete PMF	explicit probability assignment	probability table
Functional PMF	algebraic rule + normalisation	$k(x^2+1)$
Derived PMF	counting structure + probability weighting	binomial

Master insight. Counting determines *structure*; probability determines *weight*. A complete discrete model requires both.

10 Review Problems for Self-Learning

10.1 Set A: Structural fluency

1. A 5-digit code uses digits 0–9 with no repetition. Count the codes. Then explain why the answer changes if repetition is allowed.
2. In a committee of 12 students, choose a chair, vice-chair, and secretary. Explain whether this is a combination or a permutation situation before calculating.
3. Show by interpretation (not full proof) why $\binom{n}{1} = n$ and $\binom{n}{n-1} = n$.

4. A discrete random variable has PMF values $0.2, 0.3, 0.1, a$. Find a and explain the validity condition used.
5. A fair 10-sided die is rolled. What is the probability of a prime number? Then explain what counting assumption made that easy.
6. A borrower defaults with probability 0.04. Model this with a Bernoulli random variable and interpret $\mathrm{E}[X]$.
7. An insurer covers 12 independent policies, each with claim probability 0.15. Write the binomial expression for exactly 3 claims.
8. Explain why $\mathrm{P}(X \geq 1) = 1 - \mathrm{P}(X = 0)$ is often smarter than summing many terms.
9. A model is given by $\mathrm{P}(X = x) = k(2x + 1)$ for $x \in \{0, 1, 2\}$. Determine the normalising condition before solving for k.
10. Compare two distributions on support $\{0, 1, 2, 3\}$: one uniform and one heavily weighted toward 3. Without calculating exact means, which must have the larger expectation? Explain.

10.2 Set B: Deeper challenges

1. A risk committee chooses 5 members from 12 actuaries and 8 statisticians, with at least 3 actuaries required. Build the counting structure before computing anything.
2. A binomial model predicts defaults in a portfolio, but a market shock makes defaults correlated. Which BINS condition fails, and which term in the formula becomes conceptually unreliable?
3. If $X \sim \mathrm{Bin}(100, p)$, explain without calculus why the variance should be largest near $p = 0.5$.
4. Create your own real-world problem that genuinely needs a combination inside a probability formula.
5. Invent a case where a discrete uniform model might be used as a baseline stress test but would be a poor long-term predictive model.
6. Build two different PMFs on the same support with the same expected value but very different spreads. What does this teach you about variance?
7. Design a transition problem that starts as pure counting, then becomes a probability question after unequal weights are introduced.

11 Master Summary Table

Idea	Core formula	Key decision question	Risk / modelling interpretation
Fundamental Counting Principle	$n_1 n_2 \cdots n_k$	Are these successive stages?	Counts complete paths in approval, failure, or execution systems.
Factorial	$n!$	Are all places filled with distinct items?	Captures explosive growth of ordered scenarios.
Combination	$\binom{n}{r} = \frac{n!}{r!(n-r)!}$	Is it the set that matters, not the ordering?	Counts portfolios, committees, audit samples, and selected risk pools.
Concrete PMF	explicit probabilities on support	Are the probabilities already assigned?	Describes uneven weighting of outcomes from data or design.
Functional PMF	formula + normalisation	Does a symbolic rule still need scaling to sum to 1?	Encodes patterns of weighting while respecting total probability.
Discrete uniform	$1/n$ each	Are all admissible values equally likely?	Appropriate for symmetry or baseline stress assumptions.
Bernoulli	$p^x(1-p)^{1-x}$	Is there exactly one binary trial?	Models a single yes/no event such as default or survival.
Binomial	$\binom{n}{x} p^x (1-p)^{n-x}$	Are you counting successes across repeated independent Bernoulli trials?	The standard frequency model for portfolios of similar independent risks.
Expectation	$\mathrm{E}[X]$	What is the long-run average?	Guides pricing and central forecast.
Variance	$\mathrm{Var}(X)$	How spread out are possible outcomes?	Guides uncertainty buffers and reserve strength.